

PORTFOLIO CASE STUDY: RE-ENGINEERING E-COMMERCE CONVERSION FUNNELS

Platform: Google Cloud Platform (GCP)

Data Engine: Google BigQuery Studio

Data Path: insightengine-sql-analytics.raw_ecommerce.web_events

Project Type: Data Analytics Core Portfolio Case Study

1. STRATEGIC BUSINESS OVERVIEW

In digital retail environments, user experience leaks directly equate to lost capital. This case study processes raw user session clickstream data to uncover exactly how platform engagement impacts downstream profitability. By building analytical metrics around acquisition streams, transaction funnels, and customer behavior cycles, we construct an executive framework for product engineering and marketing alignment.

2. STRUCTURAL DATA SCHEMA

The operational environment ingests unstructured event arrays into a defined landing table schema:

- event_id (INTEGER): Globally unique transaction entry tracking token.
- user_id (INTEGER): Unique customer entity identifier.
- event_type (STRING): Session state boundary (page_view, add_to_cart, checkout_start, payment_info, purchase).
- event_date (TIMESTAMP): Exact millisecond logging matrix.
- product_id (INTEGER): Unique SKU reference key.
- amount (FLOAT): Explicit transactional dollar value (NULL for browsing rows).
- traffic_source (STRING): Attribution channel tracking (organic, social, paid_ads, email).

3. CORE ANALYTICAL QUESTS

QUEST 1: MARKETING ATTRIBUTION & CONVERSION ROI

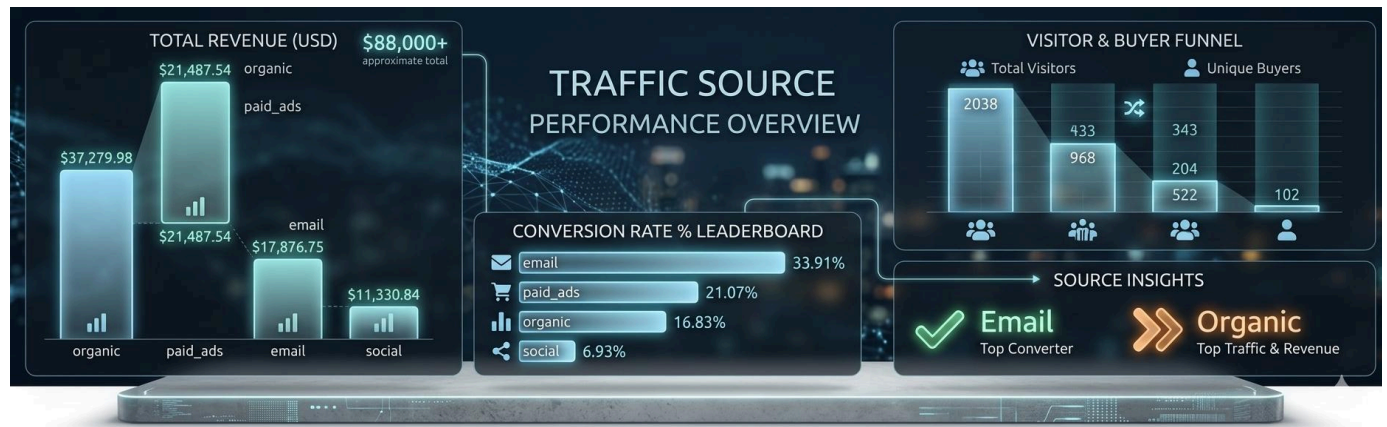
Objective: Evaluate the absolute financial contribution versus structural target traffic quality across all digital advertising streams.

```
SELECT
  traffic_source,
  COUNT(DISTINCT user_id) AS total_visitors,
  COUNT(DISTINCT CASE WHEN event_type = 'purchase' THEN user_id END) AS
unique_buyers,
  ROUND(SUM(CASE WHEN event_type = 'purchase' THEN amount ELSE 0 END), 2) AS
total_revenue_usd,
  ROUND(
```

```

    (COUNT(DISTINCT CASE WHEN event_type = 'purchase' THEN user_id END) /
COUNT(DISTINCT user_id)) * 100,
    2
) AS conversion_rate_percentage
FROM
    `insightengine-sql-analytics.raw_ecommerce.web_events`
GROUP BY
    traffic_source
ORDER BY
    total_revenue_usd DESC;

```



Executive Summary:

- High-volume, high-revenue streams indicate core brand dominance; protect and maintain baseline budgets here.
- Lower-traffic paths showing high conversion rate percentages reveal targeted, high-intent cohorts. Reallocate experimental budget here away from low-converting, expensive mass ad channels.

QUEST 2: USER CONVERSION FUNNEL LEAKAGE ANALYSIS

Objective: Calculate linear user volumes, drop-off scales, and platform abandon rates at critical milestones in the purchasing engine.

```

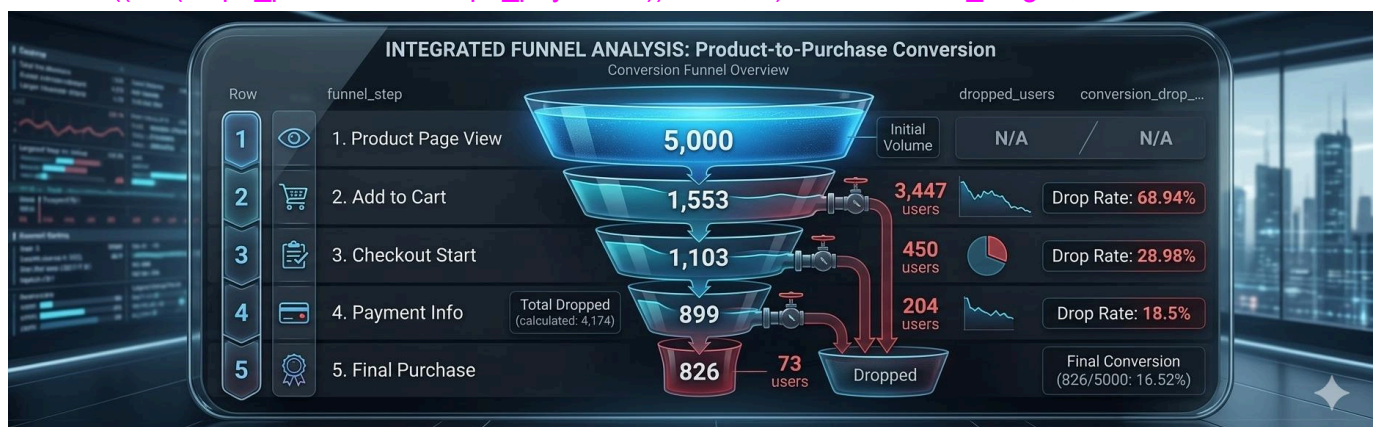
WITH funnel_stages AS (
    SELECT
        COUNT(DISTINCT CASE WHEN event_type = 'page_view' THEN user_id END) AS
step1_views,
        COUNT(DISTINCT CASE WHEN event_type = 'add_to_cart' THEN user_id END) AS
step2_carts,
        COUNT(DISTINCT CASE WHEN event_type = 'checkout_start' THEN user_id END) AS
step3_checkouts,
        COUNT(DISTINCT CASE WHEN event_type = 'payment_info' THEN user_id END) AS
step4_payments,

```

```

COUNT(DISTINCT CASE WHEN event_type = 'purchase' THEN user_id END) AS
step5_purchases
FROM
`insightengine-sql-analytics.raw_ecommerce.web_events`
)
SELECT '1. Product Page View' AS funnel_step, step1_views AS user_count, NULL AS
dropped_users, NULL AS conversion_drop_rate_pct FROM funnel_stages
UNION ALL
SELECT '2. Add to Cart', step2_carts, (step1_views - step2_carts), ROUND((1 - (step2_carts /
step1_views)) * 100, 2) FROM funnel_stages
UNION ALL
SELECT '3. Checkout Start', step3_checkouts, (step2_carts - step3_checkouts), ROUND((1 -
(step3_checkouts / step2_carts)) * 100, 2) FROM funnel_stages
UNION ALL
SELECT '4. Payment Info', step4_payments, (step3_checkouts - step4_payments), ROUND((1 -
(step4_payments / step3_checkouts)) * 100, 2) FROM funnel_stages
UNION ALL
SELECT '5. Final Purchase', step5_purchases, (step4_payments - step5_purchases),
ROUND((1 - (step5_purchases / step4_payments)) * 100, 2) FROM funnel_stages;

```



Executive Summary:

- Severe drop-off at Step 2 indicates pricing or information friction on item descriptions.
- Drop-offs occurring after Step 3 highlight transaction platform flaws, such as hidden delivery costs, lack of localized checkout types, or system rendering bottlenecks.

QUEST 3: PRODUCT PERFORMANCE & MERCHANDISING EFFICIENCY

Objective: Rank individual product IDs by raw view traffic versus successful conversion volume and total sales generation.

```

SELECT
product_id,
COUNT(CASE WHEN event_type = 'page_view' THEN 1 END) AS total_views,

```

```

COUNT(CASE WHEN event_type = 'purchase' THEN 1 END) AS total_units_sold,
ROUND(SUM(CASE WHEN event_type = 'purchase' THEN amount ELSE 0 END), 2) AS
total_revenue_usd
FROM
`insightengine-sql-analytics.raw_ecommerce.web_events`
GROUP BY
product_id
ORDER BY
total_revenue_usd DESC;

```



Executive Summary:

- High views with low sales conversions point to competitive pricing strategy issues or inaccurate stock availability flags on product pages.
- Lower view volumes with strong sales metrics reveal high-intent products that can be boosted with increased home page feature real estate or ad placements.

QUEST 4: BEHAVIORAL VELOCITY AND INTERACTION INTENSITY

Objective: Measure buyer journey length and user interaction footprints prior to final purchase execution.

[SQL CODE]

```

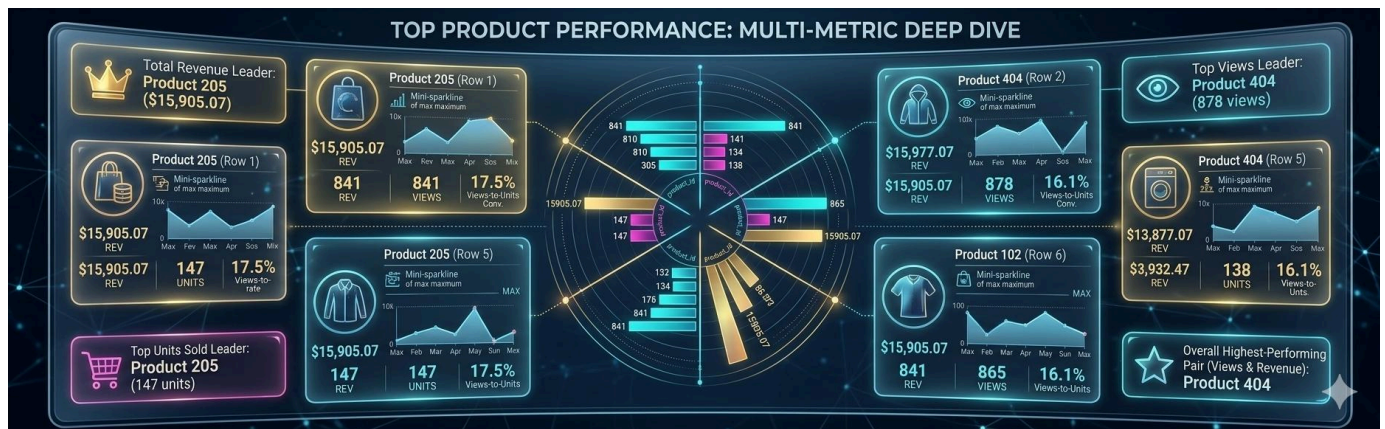
WITH buyer_journey AS (
  SELECT
    user_id,
    COUNT(event_id) AS total_touchpoints
  FROM
    `insightengine-sql-analytics.raw_ecommerce.web_events`
  WHERE
    user_id IN (
      SELECT DISTINCT user_id
      FROM `insightengine-sql-analytics.raw_ecommerce.web_events`
    )
)

```

```

WHERE event_type = 'purchase'
)
GROUP BY
user_id
)
SELECT
MIN(total_touchpoints) AS min_actions_before_purchase,
MAX(total_touchpoints) AS max_actions_before_purchase,
ROUND(AVG(total_touchpoints), 1) AS avg_actions_before_purchase
FROM
buyer_journey;

```



Executive Summary:

- If average buyer interaction intensity requires multiple distinct site touches, single-session impulse buys are low.
- Marketing must build structured multi-tier engagement workflows (like cart recovery triggers deployed at optimal hour targets) to align with standard user behavior.